# MPG No. 32.5.122 $m'\bar{3}'m'$ [ Type III, cubic ]

Table 1: Wyckoff site: `1o`, site symmetry: `m'-3'm'`

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | `[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,` |
| | | `25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]` |

Table 2: Wyckoff site: `6a`, site symmetry: `4m.m'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | `[1,2,19,20,27,28,41,42]` |
| 2 | $[-x, 0, 0]$ | `[3,4,17,18,25,26,43,44]` |
| 3 | $[0, x, 0]$ | `[9,11,14,16,34,36,37,39]` |
| 4 | $[0, -x, 0]$ | `[10,12,13,15,33,35,38,40]` |
| 5 | $[0, 0, x]$ | `[5,8,22,23,30,31,45,48]` |
| 6 | $[0, 0, -x]$ | `[6,7,21,24,29,32,46,47]` |

Table 3: Wyckoff site: `8b`, site symmetry: `.3m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, x]$ | `[1,5,9,37,41,45]` |
| 2 | $[-x, -x, x]$ | `[4,8,12,40,44,48]` |
| 3 | $[-x, x, -x]$ | `[3,7,11,39,43,47]` |
| 4 | $[x, -x, -x]$ | `[2,6,10,38,42,46]` |
| 5 | $[x, x, -x]$ | `[16,20,24,28,32,36]` |
| 6 | $[-x, -x, -x]$ | `[13,17,21,25,29,33]` |
| 7 | $[x, -x, x]$ | `[15,19,23,27,31,35]` |
| 8 | $[-x, x, x]$ | `[14,18,22,26,30,34]` |

Table 4: Wyckoff site: `12c`, site symmetry: `m'.m2`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, y, y]$ | `[1,18,26,41]` |
| 2 | $[0, -y, y]$ | `[4,19,27,44]` |
| 3 | $[0, y, -y]$ | `[3,20,28,43]` |
| 4 | $[0, -y, -y]$ | `[2,17,25,42]` |
| 5 | $[y, 0, y]$ | `[9,15,35,37]` |
| 6 | $[y, 0, -y]$ | `[10,16,36,38]` |
| 7 | $[-y, 0, y]$ | `[12,14,34,40]` |
| 8 | $[-y, 0, -y]$ | `[11,13,33,39]` |
| 9 | $[y, y, 0]$ | `[5,24,32,45]` |
| 10 | $[-y, y, 0]$ | `[7,22,30,47]` |
| 11 | $[y, -y, 0]$ | `[6,23,31,46]` |
| 12 | $[-y, -y, 0]$ | `[8,21,29,48]` |

Table 5: Wyckoff site: `24d`, site symmetry: `m'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, y, z]$ | `[1,26]` |
| 2 | $[0, -y, z]$ | `[4,27]` |
| 3 | $[0, y, -z]$ | `[3,28]` |
| 4 | $[0, -y, -z]$ | `[2,25]` |
| 5 | $[z, 0, y]$ | `[9,35]` |
| 6 | $[z, 0, -y]$ | `[10,36]` |
| 7 | $[-z, 0, y]$ | `[12,34]` |
| 8 | $[-z, 0, -y]$ | `[11,33]` |
| 9 | $[y, z, 0]$ | `[5,32]` |
| 10 | $[-y, z, 0]$ | `[7,30]` |
| 11 | $[y, -z, 0]$ | `[6,31]` |
| 12 | $[-y, -z, 0]$ | `[8,29]` |
| 13 | $[y, 0, -z]$ | `[16,38]` |
| 14 | $[-y, 0, -z]$ | `[13,39]` |
| 15 | $[y, 0, z]$ | `[15,37]` |
| 16 | $[-y, 0, z]$ | `[14,40]` |
| 17 | $[0, z, -y]$ | `[20,43]` |
| 18 | $[0, z, y]$ | `[18,41]` |
| 19 | $[0, -z, -y]$ | `[17,42]` |
| 20 | $[0, -z, y]$ | `[19,44]` |
| 21 | $[z, y, 0]$ | `[24,45]` |
| 22 | $[z, -y, 0]$ | `[23,46]` |
| 23 | $[-z, y, 0]$ | `[22,47]` |
| 24 | $[-z, -y, 0]$ | `[21,48]` |

Table 6: Wyckoff site: `24e`, site symmetry: `..m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, z]$ | `[1,37]` |
| 2 | $[-x, -x, z]$ | `[4,40]` |
| 3 | $[-x, x, -z]$ | `[3,39]` |
| 4 | $[x, -x, -z]$ | `[2,38]` |
| 5 | $[z, x, x]$ | `[9,45]` |
| 6 | $[z, -x, -x]$ | `[10,46]` |
| 7 | $[-z, -x, x]$ | `[12,48]` |
| 8 | $[-z, x, -x]$ | `[11,47]` |
| 9 | $[x, z, x]$ | `[5,41]` |
| 10 | $[-x, z, -x]$ | `[7,43]` |
| 11 | $[x, -z, -x]$ | `[6,42]` |
| 12 | $[-x, -z, x]$ | `[8,44]` |
| 13 | $[x, x, -z]$ | `[16,28]` |
| 14 | $[-x, -x, -z]$ | `[13,25]` |
| 15 | $[x, -x, z]$ | `[15,27]` |
| 16 | $[-x, x, z]$ | `[14,26]` |
| 17 | $[x, z, -x]$ | `[20,32]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 18 | $[-x, z, x]$ | [18,30] |
| 19 | $[-x, -z, -x]$ | [17,29] |
| 20 | $[x, -z, x]$ | [19,31] |
| 21 | $[z, x, -x]$ | [24,36] |
| 22 | $[z, -x, x]$ | [23,35] |
| 23 | $[-z, x, x]$ | [22,34] |
| 24 | $[-z, -x, -x]$ | [21,33] |

Table 7: Wyckoff site: 48f, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, z]$ | [1] |
| 2 | $[-x, -y, z]$ | [4] |
| 3 | $[-x, y, -z]$ | [3] |
| 4 | $[x, -y, -z]$ | [2] |
| 5 | $[z, x, y]$ | [9] |
| 6 | $[z, -x, -y]$ | [10] |
| 7 | $[-z, -x, y]$ | [12] |
| 8 | $[-z, x, -y]$ | [11] |
| 9 | $[y, z, x]$ | [5] |
| 10 | $[-y, z, -x]$ | [7] |
| 11 | $[y, -z, -x]$ | [6] |
| 12 | $[-y, -z, x]$ | [8] |
| 13 | $[y, x, -z]$ | [16] |
| 14 | $[-y, -x, -z]$ | [13] |
| 15 | $[y, -x, z]$ | [15] |
| 16 | $[-y, x, z]$ | [14] |
| 17 | $[x, z, -y]$ | [20] |
| 18 | $[-x, z, y]$ | [18] |
| 19 | $[-x, -z, -y]$ | [17] |
| 20 | $[x, -z, y]$ | [19] |
| 21 | $[z, y, -x]$ | [24] |
| 22 | $[z, -y, x]$ | [23] |
| 23 | $[-z, y, x]$ | [22] |
| 24 | $[-z, -y, -x]$ | [21] |
| 25 | $[-x, -y, -z]$ | [25] |
| 26 | $[x, y, -z]$ | [28] |
| 27 | $[x, -y, z]$ | [27] |
| 28 | $[-x, y, z]$ | [26] |
| 29 | $[-z, -x, -y]$ | [33] |
| 30 | $[-z, x, y]$ | [34] |
| 31 | $[z, x, -y]$ | [36] |
| 32 | $[z, -x, y]$ | [35] |
| 33 | $[-y, -z, -x]$ | [29] |
| 34 | $[y, -z, x]$ | [31] |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 35 | $[-y,\ z,\ x]$ | [30] |
| 36 | $[y,\ z,\ -x]$ | [32] |
| 37 | $[-y,\ -x,\ z]$ | [40] |
| 38 | $[y,\ x,\ z]$ | [37] |
| 39 | $[-y,\ x,\ -z]$ | [39] |
| 40 | $[y,\ -x,\ -z]$ | [38] |
| 41 | $[-x,\ -z,\ y]$ | [44] |
| 42 | $[x,\ -z,\ -y]$ | [42] |
| 43 | $[x,\ z,\ y]$ | [41] |
| 44 | $[-x,\ z,\ -y]$ | [43] |
| 45 | $[-z,\ -y,\ x]$ | [48] |
| 46 | $[-z,\ y,\ -x]$ | [47] |
| 47 | $[z,\ -y,\ -x]$ | [46] |
| 48 | $[z,\ y,\ x]$ | [45] |